

Math 8 Spring Break Review

Integer Operations • Ratios and Rates • Fraction Operations Name: _____

Instructions:

During the break, complete **a task each weekday** to keep your math skills sharp.

Each task includes:

- **Green** – Core Skills
- **Blue** – Proficiency Problems
- **Math Thinking Reflection** – Explain your reasoning

You may complete more than two tasks in a day, but **spacing practice over time helps your brain remember the skills better.**

This review is to be completed and handed in on: _____

Reflection Expectations

For each task, write **3-5 sentences** explaining your thinking.

Your explanation may include:

- Examples
- Diagrams
- Number lines
- Equations

The goal is to **communicate mathematical thinking clearly.**

Task 1 – Integer Operations

Evaluate the following expressions. Show your work if possible. Do the work without a calculator.

Green

1. $7 + (-4)$

2. $-9 + 13$

3. $6 - (-3)$

4. $-8 - 6$

Blue

5. $-5 + 8 - 7$

6. $10 - (-4) + (-6)$

7. $-12 + 5 - (-2)$

8. $-4 + 3(-2) + 5$

Reflection

Why does **subtracting a negative number become addition**?

Task 2 – Integer Order of Operations

Evaluate the following expressions. Show your work if possible. Do the work without a calculator.

Green

1. $5 + 2(4 - 6)$

2. $12 - 3(5 - 7)$

3. $4(6 - 8) + 9$

Blue

4. $6 + 3(4 - 7) + 2$

5. $15 - 4(3 - 6) + 1$

6. $-5 + 2(6 - 9)$

7. $8 + 3(5 - 9) - 2(3)$

Reflection

Why do we need **order of operations** in mathematics?

Task 3 - Ratios

Evaluate the following ratios.

Green

1. Simplify 6: 12
2. Simplify 8: 20
3. Write 4: 10 as a fraction in simplest form.
4. A class has **12 boys and 18 girls**.
Write the ratio boys : girls.

Blue

5. The ratio of **red to blue marbles is** 3: 5.
If there are **15 red marbles**, how many blue marbles are there?
6. The ratio of **boys to girls is** 2: 3.
If there are **30 students**, how many are boys?
7. The ratio of **cats to dogs is** 4: 7.
If there are **44 animals**, how many are dogs?
8. A paint mixture uses the ratio **3:5** (blue : yellow).
If the mixture contains **16 cups**, how many cups are yellow?

Reflection

What does a **ratio represent**?

Give a real-life example.

Task 4 – Rates and Unit Rates

Green

1. 12 cookies cost \$4.
Find the **price per cookie**.
2. A car travels **180 km in 3 hours**.
Find the speed in **km/h**.
3. A printer prints **24 pages in 6 minutes**.
Find the pages per minute.

Blue

4. A cyclist travels **45 km in 3 hours**.
How far will they travel in **5 hours**?
5. A runner travels **8 km in 40 minutes**.
How far would they travel in **1 hour**?
6. A recipe uses **2 cups sugar for 3 cups flour**.
How much sugar is needed for **12 cups flour**?
7. A rope is cut into pieces in the ratio **3:4:5**.
The rope is **36 m long**.
How long is each piece?

Reflection

Why are **unit rates useful when comparing choices**?

Task 5 – Fraction Addition & Subtraction

Evaluate each expression. Show your work and write your fraction in simplest terms.

Green

1. $\frac{3}{4} + \frac{1}{6} - \frac{1}{3}$

2. $\frac{5}{8} - \frac{1}{3} + \frac{1}{4}$

3. $\frac{2}{3} + \frac{3}{10} - \frac{1}{5}$

4. $\frac{7}{12} - \frac{1}{4} + \frac{1}{6}$

Blue

5. $\frac{3}{4} + \frac{2}{5} - \frac{1}{3} + \frac{1}{6}$

6. $\frac{5}{6} - \frac{1}{4} + \frac{2}{3} - \frac{1}{2}$

7. $\frac{7}{8} + \frac{3}{10} - \frac{1}{4} + \frac{1}{5}$

8. $\frac{5}{6} + \frac{7}{12} - \frac{3}{4} + \frac{2}{3}$

Reflection

Why do fractions need **common denominators** when adding or subtracting?

Task 6 – Fraction Multiplication

Find the product.

Green

1. $\frac{2}{3} \times \frac{3}{4} \times \frac{5}{6}$

2. $\frac{4}{5} \times \frac{2}{3} \times \frac{3}{8}$

3. $\frac{5}{6} \times \frac{1}{2} \times \frac{3}{5}$

Blue

4. $\frac{3}{4} \times \frac{5}{8} \times \frac{4}{3} \times \frac{2}{5}$

5. $\frac{6}{7} \times \frac{14}{15} \times \frac{5}{6} \times \frac{3}{4}$

9. $\frac{8}{9} \times \frac{3}{4} \times \frac{3}{2} \times \frac{5}{6}$

10. $1\frac{1}{2} \times \frac{3}{5} \times \frac{4}{3} \times \frac{2}{7}$

Reflection

What does it mean to find **a fraction of a quantity**?

Task 7 – Fraction Division

Find the quotient

Green

1. $\frac{3}{4} \div \frac{1}{2} \times \frac{3}{5}$

2. $\frac{5}{6} \div \frac{1}{3} \times \frac{1}{4}$

3. $\frac{7}{8} \div \frac{1}{4} \times \frac{2}{3}$

Blue

4. $\frac{4}{5} \div \frac{2}{3} \times \frac{3}{8} \times \frac{6}{5}$

5. $\frac{9}{10} \div \frac{3}{5} \times \frac{5}{6} \times \frac{3}{4}$

6. $\frac{7}{12} \div \frac{2}{3} \times \frac{3}{4} \times \frac{8}{5}$

7. $1\frac{3}{4} \div \frac{2}{5} \times \frac{1}{2} \times \frac{3}{4}$

Reflection

Why does **dividing by a fraction mean multiplying by its reciprocal**?

Task 8 – Mixed Fraction Operations

Evaluate the following expressions

Green

1. $1\frac{1}{2} + \frac{2}{3} - \frac{1}{4}$

2. $2\frac{1}{3} - \frac{3}{4} + \frac{1}{6}$

3. $1\frac{3}{4} + \frac{2}{5} - \frac{1}{2}$

Blue

4. $2\frac{1}{2} + 1\frac{3}{4} - \frac{2}{3} + \frac{5}{6}$

5. $3\frac{1}{3} - 1\frac{2}{5} + \frac{3}{4} - \frac{1}{6}$

6. $1\frac{1}{2} \times \frac{3}{4} + \frac{2}{3} - \frac{1}{5}$

7. $1\frac{1}{2} \times \frac{3}{4} + \frac{2}{3} - \frac{1}{5}$

Reflection

Why is it often easier to convert **mixed numbers to improper fractions**?

TASK 9 – Fractions and Ratios

Green

1. Simplify 12: 18
2. Write $\frac{2}{3}$ as a ratio.
3. Write 15: 20 in simplest form.

Blue

4. A recipe uses **2 cups sugar for 3 cups flour**.
Write the ratio sugar : flour.
5. A bag contains **8 red and 12 blue marbles**.
Write the ratio in simplest form.
6. If $\frac{2}{5}$ of a class are girls, what is the ratio girls : students?
7. A class has **30 students and $\frac{2}{3}$ are present**.
How many students are present?

Reflection

Explain how **fractions and ratios are connected**.

TASK 10 – Mixed Review

Green

1. $-6 + 4(3 - 5)$

2. $\frac{3}{4} + \frac{1}{2}$

3. Simplify 18:24

Blue

4. $-3 + 2(4 - 9) + 5$

5. $\frac{2}{3} \times \frac{3}{5}$

6. $\frac{3}{4} \div \frac{1}{2}$

7. A recipe uses **3 cups flour for 2 cups milk**.
How much flour is needed for **10 cups milk**?

Reflection

Which concept was **most challenging** for you in this review?

Explain why.